Yumu Yang

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Education

- **PhD in Physics** *2020 - Current*
  Advisor since 2022: Jorge Noronha University of Illinois Urbana-Champaign
- **BA (Hons.) in Physics and Psychology** *2016 - 2020*
  Dean's List, Psi Chi Member Colby College

Research Interests

Neutron stars compress matter to several times the density of an atomic nucleus, a regime beyond the reach of first-principle techniques and inaccessible to laboratory experiment. My research asks which properties of such matter are fixed by symmetry and by nuclear constants that have already been measured, and which genuinely depend on the model one adopts. I have shown that the bulk-viscous transport coefficients of neutron star matter are determined by the nuclear symmetry energy alone, and at saturation density by two experimentally constrained coefficients; that during a binary inspiral the matter responds elastically rather than viscously; and that the symmetry-energy expansion itself must be rebuilt from the flavor symmetry of QCD once strange particles are present. I have also developed a thermodynamically consistent method for merging equations of state, now in use within the NSF MUSES Collaboration. This work connects nuclear theory directly to gravitational-wave and X-ray observations from LIGO/Virgo and NICER, with the aim of making dense-matter predictions falsifiable rather than model-dependent.

Honors

- **Vijay R. Pandharipande Prize in Nuclear Physics** *2026*
  Competitive departmental research award ($2000) University of Illinois Urbana-Champaign
- **Mavis Future Faculty Fellow** *2026-2027*
  Fellowship supporting PhD students preparing for academic careers ($2000) University of Illinois Urbana-Champaign
- **Tau Beta Pi** *2026*
  Engineering Honor Society University of Illinois Urbana-Champaign

Experience

- **Nuclear Theory – Noronha Group** *2022 - Current*
  Research/Teaching Assistant University of Illinois Urbana-Champaign
  - Developed and maintained the holographic module for the NSF MUSES Collaboration.
  - 2024 - 2025: Organized the MUSES Collaboration Seminars.
  - Determined the bulk viscosity of nuclear matter at saturation density; calculated new bulk-viscous transport coefficients in neutron star matter; studied the effects of the nuclear equation of state on the dissipative properties of neutron star mergers/inspirals.
- **Biological Physics – Golding Lab** *2020 - 2022*
  Research/Teaching Assistant – Golding Group University of Illinois Urbana-Champaign
  - Studied cell transcription with experimental techniques such as smFISH.
- **DarkSide Collaboration (dark matter experiment)** *Summer 2019*
  Research Assistant Gran Sasso National Laboratory, Italy
  - Tested Silicon Photomultipliers in both room temperature and cryogenic conditions.
- **Colby College** *2017 - 2020*
  Research/Teaching Assistant
  - Studied autocatalytic reactions, modeled and simulated the perturbed Brusselator.
  - Built laser diodes.
  - Graded students' homework and tutored physics courses.

Technical Skills

- **Programming Languages:** C++, Python, Matlab, Mathematica.

List of Publications

C=Conference, J=Journal, P=ArXiv Preprint

[P.1] Johannes Jahan, Kevin P. Pala, **Yumu Yang** et al., "Studying the QCD Matter produced in Heavy-Ion Collisions using the MUSES Calculation Engine," arXiv: 2606.26326 [nucl-th].

[P.2] Musa Khan, Ayrton Nascimento, **Yumu Yang** et al., "Uncertainty quantification of holographic transport and energy loss for the hot and baryon-dense QGP," arXiv: 2603.20482 [nucl-th], submitted to PRD.

[J.1] Abhishek Hegade, **Yumu Yang** et al., "Dynamical tidal response of neutron stars as a probe of dense-matter properties," Phys.Rev.D 114 (2026), arXiv: 2603.26886 [gr-qc].

[J.2] **Yumu Yang** et al., "Merging multidimensional equations of state of strongly interacting matter via a statistical mixture," Phys.Rev.D 113 (2026), arXiv: 2601.07987 [nucl-th].

[J.3] Isabella Danhoni, **Yumu Yang**, Mauricio Hippert, and Jacquelyn Noronha-Hostler, "Symmetry Energy of 2+1-flavor dense quark matter from perturbative QCD," Phys.Rev.C 113 (2026), arXiv: 2510.23984 [nucl-th].

[J.4] **Yumu Yang**, Nikolas Cruz Camacho, Mauricio Hippert, and Jacquelyn Noronha-Hostler, "Symmetry-Energy Expansion with Strange Dense Matter," Phys.Rev.C 113 (2026), arXiv: 2504.18764 [nucl-th].

[J.5] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Symmetry-energy dependence of the bulk viscosity of nuclear matter," Phys. Rev. Lett. 135 (2025), arXiv: 2504.07805 [nucl-th].

[J.6] Mateus Pelicer et al., "Building Neutron Stars with the MUSES Calculation Engine," Phys. Rev. D 111 (2025), arXiv: 2502.07902 [nucl-th].

[J.7] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Far-from-equilibrium bulk-viscous transport coefficients in neutron star mergers," Phys. Rev. C 109 (2024), arXiv: 2309.01864 [nucl-th].

[J.8] Robert Bluhm and **Yumu Yang**, "Gravity with Explicit Diffeomorphism Breaking," Symmetry 13 (2021), arXiv: 2104.05879 [gr-qc].

[C.1] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Bulk viscosity dependence on the symmetry energy," Quark Matter 2025.

[C.2] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Bulk viscosity transport coefficients in neutron star mergers," Quark Matter 2023, arXiv: 2312.09364 [nucl-th].

[C.3] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Second-Order Transport Coefficients in Neutron Star Mergers," Hot Quarks 2022, arXiv: 2402.06878 [nucl-th].

List of Oral Presentations

- **Strangeness in Quark Matter 2026** — *2026*
  Connecting heavy-ion collisions to strange dense matter in neutron stars — University of California Los Angeles
- **Workshop on High Baryon Density Physics in High-Energy** — *2026*
  Nuclear matter EoS at high density region — Lawrence Berkeley National Laboratory
- **The MUSES Seminar Series** — *2026*
  Merging multi-dimensional equations of state via a statistical mixture — online
- **Nuclear Physics in Mergers - Going Beyond the Equation of State** — *2025*
  Dependence of the bulk viscosity of neutron star matter on the nuclear symmetry energy — University of Washington
- **Quark Matter 2025 (largest and most important conference in the field)** — *2025*
  Dependence of the bulk viscosity of neutron star matter on the nuclear symmetry energy — Goethe University Frankfurt
- **Midwest Theory Get-Together 2024** — *2024*
  How does the neutron star bulk viscosity depend on the symmetry energy? — Argonne National Laboratory
- **Quark Matter 2023** — *2023*
  Far-from-equilibrium transport coefficients in neutron star mergers — Hilton of the Americas, Houston
- **Hot Quarks 2022** — *2022*
  Second-order transport coefficients in neutron star mergers — Dao House, Estes Park, Colorado
- **Midwest Theory Get-Together 2022** — *2022*
  Second-order transport coefficients in neutron star mergers — Argonne National Laboratory